Soroush Shahbaznejad

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RESEARCH SUMMARY

Fourth-year Ph.D. student in Color Science at the Munsell Color Science Laboratory, Rochester Institute of Technology (RIT), working at the intersection of human visual perception, material appearance, and computer vision. My research measures how people perceive glossiness, texture, and translucency in real photographs and tests whether pretrained vision foundation models and vision-language models capture these judgments, toward perceptually calibrated control of image generation and HDR reproduction. Experienced in psychophysical experiment design, display and spectral measurement, and GPU-based deep-learning pipelines.

EDUCATION

Rochester Institute of Technology (RIT), Munsell Color Science Laboratory 2023 – Expected 2028
Ph.D. in Color Science | GPA: 3.9/4.0 | Advisor: Prof. Mekides Assefa Abebe
Relevant coursework: Visual Perception Modeling, Color Imaging, Deep Learning, Computer Vision, Human Vision

Amirkabir University of Technology 2014 – 2016
M.Sc. in Textile Sciences and Engineering
Thesis: Evaluating the Effect of Near-Gray Backgrounds on Fluorescent Samples and Whether CIECAM02 Predicts Their Appearance

University of Guilan 2010 – 2014
B.Sc. in Textile Sciences and Engineering | *Ranked 1st in the program*

PUBLICATIONS

Journal Articles

[J1] **Shahbaznejad, S.**, Abebe, M. A., Rabbanifar, A., & Murdoch, M. J. (2026). Analyzing the Impact of Color Spaces and Color Augmentation on Material Segmentation and Feature Representation. *Journal of Imaging Science and Technology*, 70(5), 050402. doi:10.2352/J.ImagingSci.Technol.2026.70.5.050402

[J2] Abebe, M. A., **Shahbaznejad, S.**, Rabbanifar, A., & Fedorovskaya, E. (2025). Unveiling the Role of Color in Skin Segmentation: Analysis of Augmentation Techniques and Color Spaces. *Journal of Imaging Science and Technology*, 69(5), 050504. doi:10.2352/J.ImagingSci.Technol.2025.69.5.050504

[J3] **Shahbaznejad, S.**, & Amirshahi, S. H. (2021). Does CIECAM02 Color Appearance Model Work Well for Fluorescent Samples? *Color Research & Application*, 46(5), 923–928. doi:10.1002/col.22661

Conference Papers

[C1] Zhao, S., **Shahbaznejad, S.**, Fedorovskaya, E., & Abebe, M. A. (2026). Assessing the Influence of Color on Gloss Perception. *Colour and Visual Computing Symposium (CVCS)*.

[C2] Rabbanifar, A., **Shahbaznejad, S.**, Fairchild, M. D., & Abebe, M. A. (2026). Rendering Emissive Colors in HDR Scenes: The Role of Reference-White Placement and Highlight Tone Mapping. *IS&T Electronic Imaging*, 38, 1–7.

Forthcoming and in Preparation

[F1] **Shahbaznejad, S.**, Rabbanifar, A., Murdoch, M. J., & Abebe, M. A. Do Pretrained Vision Models Reflect Human Material Judgments? To be presented at *IS&T Electronic Imaging 2027*.

[F2] **Shahbaznejad, S.**, et al. Frozen Pretrained Vision Representations and Human Judgments of Material Appearance in Natural Photographs. Manuscript complete; submission to *Computer Vision and Image Understanding* planned for November 2026.

RESEARCH & PROFESSIONAL EXPERIENCE

Doctoral Researcher, Munsell Color Science Laboratory Rochester Institute of Technology, 2023 – Present

- Designed and ran a psychophysical study of glossiness, texture, and translucency: 300 material regions from

OpenSurfaces photographs, 35 observers, and 3,770 ratings, with stimulus-mean reliability of 0.91–0.97.
- Built the web-based experiment platform for collecting perceptual ratings of segmented material regions.
- Benchmarked 14 frozen pretrained vision models (including DINOv2, CLIP, MAE, iBOT, MoCo v3, SAM 2, Stable Diffusion VAE, Marigold, and Swin-L) against human ratings, using photograph-grouped nested cross-validation, a training-data leakage audit, material-category controls, and bootstrap inference.
- Showed that frozen features predict appearance ratings largely through material identity, with a small but reliable within-category signal; this result motivates perceptually supervised adaptation.
- Built GPU feature-extraction and analysis pipelines on RIT's SPORC high-performance computing cluster (SLURM job arrays with per-model checkpointing).
- Contributed to psychophysical studies of HDR emissive-color rendering and highlight tone mapping, and of color effects on perceived gloss.

Teaching Assistant – Principles of Color Science Rochester Institute of Technology, Fall 2025

- Supported lectures and laboratories for the graduate course taught by Prof. Michael J. Murdoch, covering instrumentation, radiometry, photometry, and spectral color measurement.

Teaching Assistant – Color Physics Amirkabir University of Technology, Feb 2023 – Jul 2023

- Led laboratory sessions on radiometry, color measurement, and material appearance; demonstrated spectroradiometer and spectrophotometer calibration; graded laboratory reports.

R&D Specialist – Color Reproducibility Nikriss Co., Feb 2020 – Oct 2021

- Improved fabric color reproducibility through spectral measurement, computational color modeling, and iterative experimental validation.
- Developed MATLAB and Python workflows for color-difference evaluation and color-calibration procedures, and worked with manufacturing and design teams to reduce production color deviations.

CONFERENCE PRESENTATIONS

- **IS&T Electronic Imaging 2027** (upcoming): Do Pretrained Vision Models Reflect Human Material Judgments?
- **IS&T Electronic Imaging 2026** (oral): Color spaces, color augmentation, and material segmentation.
- **IS&T Electronic Imaging 2025**: Color spaces and color augmentation for semantic skin segmentation.

HONORS & SCHOLARSHIPS

- **Danny Rich Memorial Scholarship research project**, Colour and Imaging Council (CIC), 2026: contributor to a funded Munsell Color Science Laboratory project on perceptually guided HDR highlight rendering and emissive color appearance.
- **Ranked 1st** in the B.Sc. program in Textile Sciences and Engineering, University of Guilan, 2014.

PROFESSIONAL SERVICE

- **Peer reviewer**, *Color Research & Application* (Wiley): manuscripts in color science, color imaging, and visual perception.

TECHNICAL SKILLS

Programming: Python, PyTorch, MATLAB, NumPy, pandas, scikit-learn, OpenCV, Git, SLURM/HPC, LaTeX
Machine learning and vision: vision transformers and foundation models (DINOv2, CLIP, SAM 2), self-supervised representations, semantic segmentation, linear and kernel probing, representational similarity analysis, clustering (GMM, HDBSCAN), PCA, UMAP
Psychophysics and statistics: experiment design, rating and comparison tasks, reliability and noise-ceiling analysis, mixed-effects models, bootstrap inference, multiple-comparison control
Color science and imaging: color appearance models, camera characterization, HDR imaging and tone mapping, spectral imaging, color calibration and management, photometry
Instrumentation: spectroradiometers (Konica Minolta CS-2000), spectrophotometers, monochromators, colorimeters (i1Display Pro), HDR displays